

MARCUS J. HOLLOWAY

marcus.holloway@protonmail.com | (415) 555-0182 | linkedin.com/in/marcusholloway | github.com/mjholloway
San Francisco, CA | Authorized to work in the US (US Citizen)

SUMMARY

Senior backend engineer with 10 years of experience designing, building, and operating distributed systems at scale. Deep expertise in service-oriented architectures, consensus protocols, and high-throughput data pipelines. Shipped infrastructure that handled tens of billions of requests per day at FAANG-adjacent companies. Comfortable owning systems end-to-end: from RFC and capacity planning through on-call rotation and post-incident review. Strong opinions, loosely held, on idempotency, backpressure, and the cost of premature abstraction.

WORK EXPERIENCE

Stripe, Inc. — Staff Software Engineer (L6)

San Francisco, CA | March 2022 – Present

Payments Orchestration Platform team. Tech lead for the merchant settlement reconciliation pipeline.

- Designed and shipped a replacement reconciliation system that processes 4.2B ledger entries per day, replacing a legacy Hadoop batch job with a streaming pipeline on Kafka and Flink. Reduced end-to-end reconciliation latency from 14 hours to 38 minutes (p99) and cut compute cost by 61% (~\$2.3M/year).
- Led the migration of 17 microservices from a monolithic API gateway to an Envoy-based service mesh. Defined the SLO framework (availability, latency, error budget burn rate) used across the org; published internal RFC adopted by 9 teams in the first quarter.
- Built a deterministic replay framework for payment workflows that reproduces production bugs from sampled traces with 99.2% fidelity. Reduced mean time to resolution on P0 payment bugs from 6.5 hours to 47 minutes.
- Owned the on-call rotation for a Tier-1 service surface (Stripe Terminal backend). Maintained 99.987% availability over 18 consecutive months across 3 quarterly SLO reviews.
- Mentored 4 engineers through promotion to senior; authored the team's distributed-systems interview rubric now used by 30+ interviewers.

Datadog — Senior Software Engineer

New York, NY | August 2019 – February 2022

Metrics ingestion platform ("Longtail" team). Owned the tail-aggregation service for high-cardinality customer metrics.

- Re-architected the metrics tail-aggregation pipeline to use a custom sharded RocksDB store with consistent hashing across 2,400 ingestion nodes. Increased sustained write throughput from 1.1M points/sec to 8.4M points/sec on the same hardware footprint.
- Designed and implemented the cross-region failover protocol for the metrics backend, using Raft over gRPC with explicit lease-based leader election. Cut regional failover time from 11 minutes to 19 seconds and eliminated 3 classes of split-brain incidents observed in the prior system.
- Shipped a query-side caching layer (Redis + LFU eviction with adaptive TTLs) that reduced p95 dashboard load times by 73% for customers with >10K active series. Drove \$4.1M in incremental ARR per the FY21 retention analysis.
- Co-led the company's adoption of OpenTelemetry for trace ingestion; authored the auto-instrumentation guide that became required reading for new backend hires.

Uber Technologies — Software Engineer II

San Francisco, CA | July 2017 – July 2019

Real-time pricing and dispatch infrastructure (Surge team). Focused on the geospatial indexing layer for the rider-driver matching system.

- Rebuilt the surge-pricing grid index from a quadtree-based PostgreSQL implementation to a custom H3-based in-memory index, reducing per-query p99 latency from 38ms to 2.4ms at the 95th percentile. Sustained through the 2018 New Year's Eve peak of 19M rides globally.
- Designed the backpressure protocol used between the dispatch service and downstream trip-state stores; eliminated a class of cascading failures that had caused 4 P0 incidents in the prior 6 months.
- Owned the schema migration tooling used by the team to evolve Redis cluster state across rolling deploys without dropping in-flight requests. Used in production for 47 zero-downtime migrations.
- Presented an internal talk on "Time, Clocks, and the Ordering Problem in Distributed Dispatch" that was mandatory viewing for new engineers on the marketplace org for two quarters.

Amazon Web Services — Software Development Engineer

Seattle, WA | August 2014 – June 2017

EC2 control plane, Instance State Management team. Owned the EBS-backed instance launch path.

- Shipped the migration of instance-launch state from a single-region MySQL cluster (with cross-region async replication) to a per-AZ DynamoDB-backed state machine. Reduced launch failures due to state-store contention by 94% and cut the median launch path by 280ms.
- Designed and implemented the optimistic concurrency control scheme for the instance-terminate API, using conditional writes plus a tombstone TTL. Eliminated the "zombie instance" class of bugs that had generated 600+ customer support tickets in the preceding year.
- Authored the team's runbook for the 2017 s2n TLS rotation across the EC2 control plane, completed 11 days ahead of the corporate security deadline.
- Received the Q4 2016 "Bar Raiser" award for technical interviewing; conducted 60+ on-site loops.

PROJECTS

tinroof — A pure-Go implementation of the Raft consensus algorithm with a built-in linearizable KV store. Used as a teaching tool at two Bay Area meetups. github.com/mjholloway/tinroof — 1.4k GitHub stars, adopted by a YC W23 startup as their config-store backbone.

obscura — A Prometheus-compatible remote-write proxy that batches and compacts metrics before forwarding, with a sidecar mode for Kubernetes pods. Reduced egress costs by 44% for the maintainer's homelab cluster of 32 nodes. Documented at obscura.dev with a published benchmark suite.

levee — An open-source circuit-breaker library for Go with adaptive thresholds based on a rolling EWMA of error rates and explicit half-open concurrency limits. Vended by 11 OSS projects (including two CNCF incubating projects). Available at github.com/mjholloway/levee — 2.1k stars, 47 contributors.

EDUCATION

University of California, Berkeley

B.S., Electrical Engineering and Computer Science | GPA 3.82/4.00 | Graduated May 2014

Relevant coursework: Distributed Systems (CS 162), Operating Systems (CS 162 grad section), Algorithms (CS 170), Databases (CS 186).

Senior thesis: "Practical Tradeoffs in Eventually-Consistent Counters" — advised by Prof. Ion Stoica.

SKILLS

Languages: Go (primary, 8+ years production), Python, Rust (working proficiency), Java, TypeScript

Distributed systems: Raft, Paxos, consistent hashing, gossip protocols, CRDTs, vector clocks, leaderless replication,

quorum systems, two-phase commit, saga pattern

Storage: PostgreSQL, MySQL, DynamoDB, RocksDB, Cassandra, Redis, Kafka, Pulsar, S3

Infra & orchestration: Kubernetes (operated 1.2k-node clusters), Envoy/Istio, Terraform, Ansible, Helm, Argo, Linkerd

Observability: Prometheus, Grafana, OpenTelemetry, Jaeger, Honeycomb, Datadog (self-explanatory)

Protocol & RPC: gRPC, Thrift, Protocol Buffers, FlatBuffers, HTTP/2, QUIC

CI/CD: GitHub Actions, Buildkite, Spinnaker, Prow

AWS: EC2, S3, RDS, DynamoDB, Kinesis, Lambda, IAM, VPC networking — Solutions Architect Associate certified (2021, since lapsed)

Other: Linux internals (cgroups, namespaces, eBPF beginner), capacity planning, chaos engineering (chaos-mesh, Gremlin), postmortem culture, technical writing, hiring & interviewing, mentoring